

math.mjs

NodeJS Async JS

Built in modules

- callback based/ promise based

http

- http server

express

1. Event Loop (<https://event-loop-explorer.vercel.app/>)

- Runtime environment for JS provides the implemetation of this Event loop
- Node as a runtime environment provides the implementation of this Evenloop which is called as LibuV

- Javascript is single threaded

Call stack

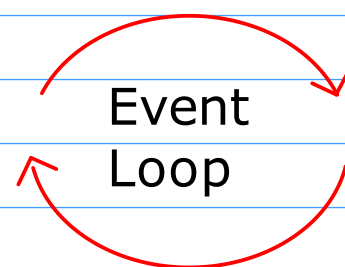
functions created on main thread will be executed first

if a function is having a callback using setTimeout it gets pushed on task queue

if a function is having a callback using promises it gets pushed on micro task queue

MicroTask Queue

callbacks created in promises



Task Queue

callback created using setTimeout

o/p

Program started

Task1 Started

Task2 Started

Task3 Started

Task3 in progress

Task3 Finished

Program Ended

Task2 in progress

Task2 Finished

Task1 in progress

Task1 Finished

How the asynchronous Behaviour works in JS...

- by the help of Event loop

Server Applications

- Web Server (live server)
- Database Server (mysql server)
- Backend Server (on Node platform we will create backend server)

- Http Methods

Postman-> API Testing tool
- where the the API can be tested for the methods like
POST, PUT, DELETE,HEAD, OPTIONS

GET
POST
PUT
DELETE

REST API

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API ->	Method	Path	
	GET	/user	user
	POST	/user	product
	PUT	/user	order
	DELETE	/user	
	GET	/product	
	POST	/product	
	PUT	/product	
	DELETE	/product	

nodemon node package manager
- node monitor (npm)

npm install -g nodemon
- it will install the nodemon globally in the device

nodemon --version

		Lab Work
/user	/product	methods
		path
		postman
		complete the 5.httpServer